

Enterprise GIS Strategy

A business-oriented plan for developing an enterprise geospatial capability

Anthony J. Sirna, GISP

Sanitized public edition

Purpose

This strategy defines how an organization can move from fragmented, inconsistently managed geospatial information to an enterprise GIS capability that supports project delivery, customer service, and data-informed decisions.

What this sample demonstrates

- Alignment of GIS investments with business goals and operational needs
- Structured assessment of data, technology, delivery processes, workforce, and governance
- Definition of a practical future state and the gaps that must be closed
- Prioritization of initiatives according to business value and ease of implementation
- Connection of strategy, governance, program management, and measurable outcomes

This edition removes organization names, locations, internal systems, budget figures, operational vulnerabilities, and customer identifiers while preserving the strategic method and decision framework.

Business Goals and Challenges

The organization needs a GIS program that improves access to reliable information, supports changing mission requirements, and creates repeatable services for internal teams and customers.

Business goals

- Maintain mission effectiveness in a resource-constrained environment.
- Provide scalable support for sustained and emerging operational requirements.
- Improve collaboration between project teams and customers.
- Increase efficiency, productivity, and growth in project delivery.

Business challenges

- Teams cannot consistently use geospatial data to answer basic questions or establish situational awareness.
- Datasets use inconsistent schemas and cannot support organization-wide comparison.
- Project deliverables and data-management practices vary by team and contract.
- Limited staffing and immature governance constrain program development.
- The organization lacks a defined portfolio of geospatial products and services.

Strategic conclusion

The organization requires a streamlined and flexible GIS program that makes authoritative data discoverable, reusable, and suitable for analysis while concentrating investment on capabilities with clear business value.

Enterprise GIS Assessment

The assessment examines the five operating dimensions that determine whether GIS can function as an enterprise capability.

Dimension	Current condition	Business consequence
Data and management	Data is distributed across project folders, hosted layers, drawings, and local holdings. Metadata, schemas, and lifecycle controls vary.	Staff spend time locating and rebuilding information. Data cannot reliably support broad analysis.
Technology	Enterprise GIS capabilities are available, but content architecture and access pathways remain underdeveloped.	Existing investments do not yet provide a consistent user experience or shared operating picture.
Applications and delivery	Most GIS support begins through informal requests and is delivered as one-time products.	Work is difficult to scale, repeat, measure, and sustain.
Workforce and culture	A small specialist workforce balances project delivery, field support, and program development.	The program depends heavily on limited staff capacity and has little redundancy.
Governance	Decision rights, stewardship, standards, and program funding are not fully institutionalized.	Priorities compete, quality varies, and long-term maintenance receives inconsistent attention.

The assessment indicates that technology alone will not solve the problem. Sustainable improvement requires coordinated changes to data management, delivery processes, workforce capacity, and governance.

Mission Vision and Guiding Principles

Mission

Provide geospatial data, tools, and services that help internal teams and customers perform their work and make better decisions.

Vision

Establish an enterprise GIS program that makes trusted geospatial information accessible and useful for project delivery, planning, analysis, and customer support.

Guiding principles

- Invest only when a capability satisfies a defined requirement, business need, or established practice.
- Prioritize initiatives with high business value and a realistic path to implementation.
- Treat geospatial information as an organizational asset with assigned ownership and lifecycle controls.
- Use commercial software and shared enterprise services where they reduce cost and maintenance burden.
- Design capabilities around user workflows and measurable outcomes.

Value proposition

The program creates value by reducing duplicated effort, improving data quality and access, supporting more consistent project delivery, and enabling analysis that cannot occur when information remains fragmented.

Target Enterprise GIS Capability

Dimension	Target condition
Data and management	An authoritative repository contains planning, real property, facility, infrastructure, utility, environmental, survey, and project information using defined schemas, metadata, stewardship, and update procedures.
Technology	An approved enterprise GIS environment provides controlled access through a central landing page, services, web maps, applications, and documented interfaces.
Applications and delivery	Standard initiation, requirements, data acquisition, production, acceptance, publication, and maintenance processes support repeatable services.
Workforce and culture	A coordinated team combines program leadership, technical specialists, project staff, and surge support. Role-based training supports adoption and succession.
Governance	Leadership-approved policy defines decision rights, priorities, funding, stewardship, access, quality, and lifecycle management.

The future state supports a progression from basic discovery and mapping to diagnostic analysis, forecasting, and documented recommendations. Advancement depends on data fitness and demonstrated business demand.

Priority Work Required to Reach the Future State

Authoritative data

- Define common schemas, metadata requirements, and acceptance criteria.
- Migrate suitable archived holdings and document provenance.
- Establish sustained feeds and update procedures for priority business data.

Enterprise delivery

- Create a content-management strategy and a central access point.
- Develop repeatable application and service patterns based on documented requirements.
- Establish a project-initiation package for imagery, maps, vector data, schemas, and data-management planning.

Workforce

- Define roles, education plans, and surge-support mechanisms.
- Plan for staffing growth as service demand becomes predictable.

Governance

- Approve decision rights, stewardship, quality controls, and publication standards.
- Establish a sustainable funding and cost-sharing approach.

Investment and Scaling Logic

Enterprise GIS investments should be matched to both expected business value and the organization's ability to launch and sustain the capability.

Technology acquisition strategy

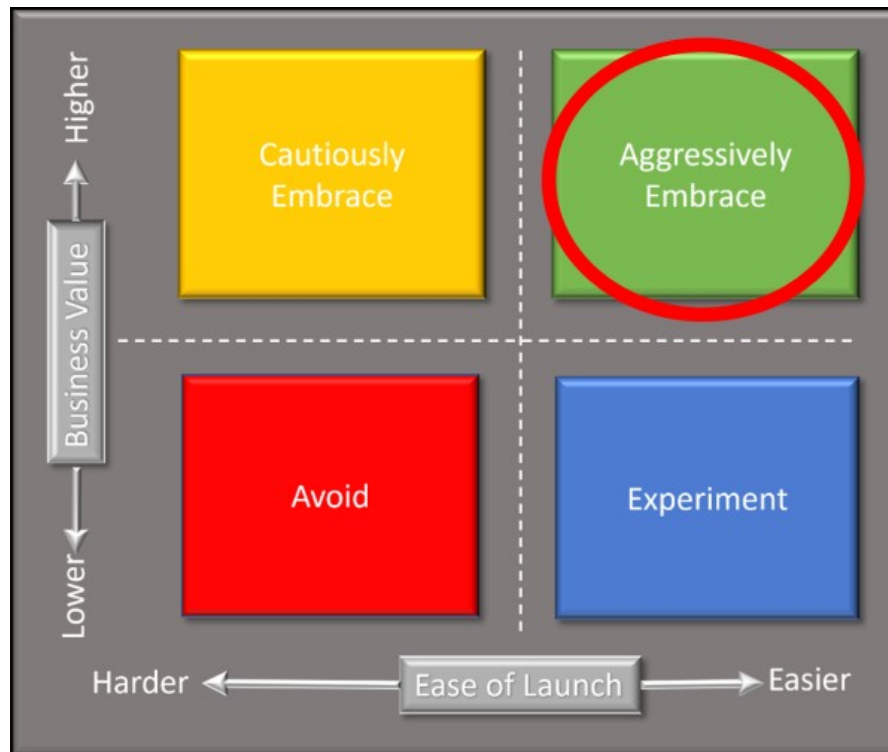


Figure 1. High-value capabilities with a practical path to implementation are the strongest candidates for aggressive adoption.

Geospatial strategy archetypes

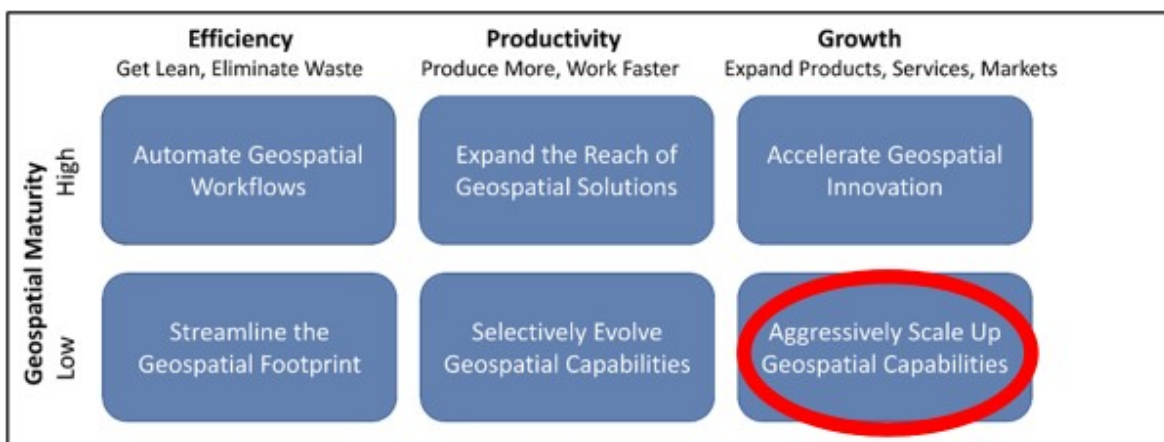


Figure 2. Strategy choices vary by geospatial maturity and the primary business objective: efficiency, productivity, or growth.

Strategic Roadmap

Phase	Focus	Representative outputs
1 Establish	Confirm priorities, governance, and architecture.	Program charter, governance framework, portfolio criteria, content strategy, stewardship assignments
2 Consolidate	Inventory, assess, standardize, and migrate priority data.	Authoritative datasets, metadata, quality findings, migration records, data dictionaries
3 Deliver	Release high-value applications and repeatable services.	Central access page, project-initiation support, viewers, dashboards, data services
4 Expand	Add sustained feeds, field collection, and customer-facing services.	Automated updates, managed data services, training, support model
5 Optimize	Measure use, value, quality, and capacity, then adjust investments.	Performance reviews, improvement backlog, staffing case, updated roadmap

Investment decisions should use a transparent portfolio method that considers expected value, implementation effort, data readiness, maintenance burden, risk, and stakeholder commitment.

Evidence of a Successful Strategy

The program should measure whether the organization has converted GIS investment into dependable capability and business value.

Outcome	Evidence
Authoritative information	Priority datasets have assigned owners, documented sources, current metadata, and defined update schedules.
Discoverability and access	Authorized users can locate and obtain relevant information through a consistent access pathway.
Reuse and efficiency	Project teams reuse trusted data and standard workflows instead of recreating one-time datasets and products.
Quality and accountability	Quality findings have owners, resolution records, and documented acceptance decisions.
Adoption	Users incorporate enterprise services into routine project and business processes.
Customer value	Customers use managed data and applications to support asset, facility, project, and environmental decisions.
Program sustainability	Leadership can connect workload, service demand, staffing, funding, and measurable outcomes.

The strategy should be reviewed as business priorities, available data, technology, and staffing change. The roadmap should remain specific enough to guide investment while allowing the program to respond to new requirements.